

Zachary Ryan McCaw

Curriculum Vitae

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Education

STANFORD UNIVERSITY

01/2021 – 06/2022

GRADUATE CERTIFICATE IN ARTIFICIAL INTELLIGENCE

- COURSEWORK: Computer Vision, Deep Learning, Reinforcement Learning.
- GPA: 4.05 of 4.00.

HARVARD UNIVERSITY

08/2014 – 05/2019

PH.D. IN BIOSTATISTICS, A.M. IN BIOSTATISTICS

- DISSERTATION: Transformation and multivariate methods for improving power in GWAS.
 - Studied operating characteristics of the rank-based inverse normal transformation for genome-wide association studies of quantitative traits. [17]
 - Developed multivariate regression methodology for leveraging a correlated surrogate outcome to improve inference on a partially missing target outcome. [13, 10]
- ADVISOR: Xihong Lin, Ph.D.
- COMMITTEE: Martin Aryee, Ph.D. and Jeffrey Miller, Ph.D.
- GPA: 3.93 of 4.00.

UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

08/2009 – 05/2013

B.S.P.H. IN BIOSTATISTICS, B.S. IN QUANTITATIVE BIOLOGY

- Graduated with highest distinction.
- GPA: 4.00 of 4.00; Dean's List: 8 of 8 Semesters; Phi Beta Kappa honors society.

Technical Experience

- **AI & ML:** Computer vision, clinical ML (survival and risk prediction), deep learning, representation learning, self-supervised learning, time-series forecasting.
- **Statistical Genetics:** Genome-wide association studies, fine-mapping, Mendelian randomization, polygenic scoring, rare-variant association testing.
- **Biostatistics & Causal Inference:** Causal inference, clinical trials, longitudinal and multivariate analysis, meta-analysis, regression modeling, survival analysis.
- **Software & Programming:** Python (PyTorch, TensorFlow), R (tidyverse), C++/Rcpp; Git; Cloud computing (AWS, GCP); Docker; Reproducible research, package development.

Professional Experience

Voleon

01/2025 – 03/2026

SENIOR MEMBER OF RESEARCH STAFF

- PROJECTS
 - Designed, implemented, and evaluated attention-based models for multivariate time series forecasting.
 - Explored multiple paradigms for representation learning from multivariate feature time series to support downstream forecasting models.

INSITRO

09/2021 – 01/2025

STAFF MACHINE LEARNING SCIENTIST: 04/23 – 01/25

SENIOR MACHINE LEARNING SCIENTIST: 09/21 – 03/23

- DEPARTMENT: Statistical Genetics and Clinical Machine Learning
- TEAM LEADS: Colm O'Dushlaine, Ph.D. and Thomas Soare, Ph.D.
- PROJECTS
 - Developed and trained machine learning models for representation learning from multiple clinical imaging modalities, including diffusion MRI, OCT, and histopathology; used the resulting embeddings for genetic discovery, risk stratification, and treatment-response prediction.
 - Led the 4-member Genetic Discovery team within the Oncology program, which focused on identifying rare germline variants associated with imaging-derived phenotypes.
 - Co-developed a framework (EmbedGEM) for evaluating the utility of embeddings for genetic discovery based on the twin objectives of association strength and clinical relevance.
 - Led development of the rare coding variant allelic series test ([COAST](#)) to identify candidate therapeutic targets that have a dose-response relationship with a target phenotype.
- PUBLICATIONS
 - Causal considerations can determine the utility of machine learning assisted GWAS. [\[2\]](#)
 - Extending the coding-variant allelic series test to summary statistics. [\[3\]](#)
 - EmbedGEM: a framework to evaluate the utility of embeddings for genetic discovery. [\[9\]](#)
 - Multi-artifact detection and filtering in digital pathology using intrinsic image properties. [\[3\]](#)
 - Machine learning enabled prediction of digital biomarkers from whole slide histopathology images. [\[5\]](#)
 - Pitfalls in performing genome-wide association studies on ratio traits. [\[5\]](#)
 - An allelic series rare-variant test for candidate gene discovery. [\[11\]](#)

GOOGLE

09/2019 – 09/2021

DATA SCIENTIST

- DEPARTMENT: Genomic Medicine Team and DevIntel Data Science Team

- **TEAM LEADS:** Babak Alipanahi, Ph.D. and Cory McLean, Ph.D. (Genomic Medicine); Heng Liu, Ph.D. (Core Developer)
- **PROJECTS**
 - Collaborated with the Genomic Medicine team on multiple projects targeting genetic discovery from machine learning-derived phenotypes, including in COPD and glaucoma.
 - Co-developed DeepNull, a machine learning-based approach to correct for non-linear covariate effects in GWAS and polygenic risk prediction.
 - Developed and implemented tools for GWAS analysis, including fine-mapping, locus formation, replication analysis, and winner’s curse correction.
 - Within Core Developer, developed and implemented methodology for estimating average causal effects from observational longitudinal data.
- **PUBLICATIONS**
 - Applying multimodal AI to physiological waveforms improves genetic prediction of cardiovascular traits. [4]
 - Unsupervised representation learning improves genomic discovery for lung function and respiratory disease prediction. [8]
 - High-throughput ML-guided design of diverse single-domain antibodies against SARS-CoV-2. [6]
 - Inference of chronic obstructive pulmonary disease with deep learning on raw spirograms identifies new genetic loci and improves risk models. [12]
 - DeepNull: Modeling non-linear covariate effects improves phenotype prediction and association power. [14]
 - Large-scale machine learning-based phenotyping significantly improves genomic discovery for optic nerve head morphology. [16]

BROAD INSTITUTE

06/2019 – 09/2019

VISITING SCIENTIST

- **DEPARTMENT:** Medical and Population Genetics.
- **PRINCIPAL INVESTIGATOR:** Hilary Finucane, Ph.D.
- **PROJECT:** Developed a cross-population extension of the sum of single effects (SuSiE) fine-mapping model for distinguishing shared and population-specific genetic effects.
- **PUBLICATION:** MultiSuSiE improves multi-ancestry fine-mapping in All of Us whole-genome sequencing data. [1]

Publications

Statistical Genetics

- [1] J Rossen et al. “MultiSuSiE improves multi-ancestry fine-mapping in All of Us whole-genome sequencing data”. In: *Nature Genetics* (Jan. 2026). DOI: [10.1038/s41588-025-02450-5](https://doi.org/10.1038/s41588-025-02450-5).

- [2] S Mukherjee et al. “Causal considerations can determine the utility of machine learning assisted GWAS”. In: *Proceedings of the sixth Conference on Health, Inference, and Learning*. Vol. 287. Proceedings of Machine Learning Research. 2025, pp. 179–193. URL: <https://proceedings.mlr.press/v287/mukherjee25a.html>.
- [3] ZR McCaw et al. “A Scalable Framework for Identifying Allelic Series from Summary Statistics”. In: *American Journal of Human Genetics* 112.11 (Nov. 2025), pp. 2772–2788. DOI: [10.1016/j.ajhg.2025.09.012](https://doi.org/10.1016/j.ajhg.2025.09.012).
- [4] Y Zhou et al. “Applying multimodal AI to physiological waveforms improves genetic prediction of cardiovascular traits”. In: *American Journal of Human Genetics* (July 2025). DOI: [10.1016/j.ajhg.2025.05.015](https://doi.org/10.1016/j.ajhg.2025.05.015).
- [5] ZR McCaw et al. “Pitfalls in performing genome-wide association studies on ratio traits”. In: *Human Genetics and Genomics Advances* (Apr. 2025). DOI: [10.1016/j.xhgg.2025.100406](https://doi.org/10.1016/j.xhgg.2025.100406).
- [6] X Li, H Chen, MS Selvaraj, et al. “A statistical framework for powerful multi-trait rare variant analysis in large-scale whole-genome sequencing studies”. In: *Nature Computational Science* (Feb. 2025). DOI: [10.1038/s43588-024-00764-8](https://doi.org/10.1038/s43588-024-00764-8).
- [7] R Sun, ZR McCaw, and X Lin. “Testing a Large Number of Composite Null Hypotheses Using Conditionally Symmetric Multidimensional Gaussian Mixtures in Genome-Wide Studies”. In: *Journal of the American Statistical Association* (Oct. 2024). DOI: [10.1080/01621459.2024.2422124](https://doi.org/10.1080/01621459.2024.2422124).
- [8] T Yun et al. “Unsupervised representation learning improves genomic discovery for lung function and respiratory disease prediction”. In: *Nature Genetics* (July 2024). DOI: [10.1038/s41588-024-01831-6](https://doi.org/10.1038/s41588-024-01831-6).
- [9] S Mukherjee et al. “EmbedGEM: A framework to evaluate the utility of embeddings for genetic discovery”. In: *Bioinformatics Advances* 4.1 (July 2024), vbae135. DOI: [10.1093/bioadv/vbae135](https://doi.org/10.1093/bioadv/vbae135).
- [10] ZR McCaw et al. “Synthetic surrogates improve power for genome-wide association studies of partially missing phenotypes in population biobanks”. In: *Nature Genetics* (June 2024). DOI: [10.1038/s41588-024-01793-9](https://doi.org/10.1038/s41588-024-01793-9).
- [11] ZR McCaw et al. “An allelic-series rare-variant association test for candidate-gene discovery”. In: *American Journal of Human Genetics* 110.8 (July 2023), pp. 1330–1342. DOI: [10.1016/j.ajhg.2023.07.001](https://doi.org/10.1016/j.ajhg.2023.07.001).
- [12] J Cosentino et al. “Inference of chronic obstructive pulmonary disease with deep learning on raw spirograms identifies new genetic loci and improves risk models”. In: *Nature Genetics* (Apr. 2023). DOI: [10.1038/s41588-023-01372-4](https://doi.org/10.1038/s41588-023-01372-4).
- [13] ZR McCaw et al. “Leveraging a surrogate outcome to improve inference on a partially missing target outcome”. In: *Biometrics* (Feb. 2022). DOI: [10.1111/biom.13629](https://doi.org/10.1111/biom.13629).
- [14] ZR McCaw et al. “DeepNull: Modeling non-linear covariate effects improves phenotype prediction and association power”. In: *Nature Communications* 13.1 (Jan. 2022), p. 241. DOI: [10.1038/s41467-021-27930-0](https://doi.org/10.1038/s41467-021-27930-0).

- [15] H Julienne et al. “Multitrait GWAS to connect disease variants and biological mechanisms”. In: *PLoS Genetics* 17.8 (Aug. 2021), e1009713. DOI: [10.1371/journal.pgen.1009713](https://doi.org/10.1371/journal.pgen.1009713).
- [16] B Alipanahi et al. “Large-scale machine learning-based phenotyping significantly improves genomic discovery for optic nerve head morphology”. In: *American Journal of Human Genetics* (May 2021). DOI: [10.1016/j.ajhg.2021.05.004](https://doi.org/10.1016/j.ajhg.2021.05.004).
- [17] ZR McCaw et al. “Operating Characteristics of the Rank-Based Inverse Normal Transformation for Quantitative Trait Analysis in Genome-Wide Association Studies”. In: *Biometrics* (Dec. 2019). DOI: [10.1111/biom.13214](https://doi.org/10.1111/biom.13214).

Applied Machine Learning

- [1] J Gao et al. “What Is Fair? Defining Fairness in Machine Learning for Health”. In: *Stat Med* 44.20-22 (Sept. 2025), e70234. DOI: [10.1002/sim.70234](https://doi.org/10.1002/sim.70234).
- [2] J Gronsbell et al. “Another look at inference after prediction”. In: *arXiv* (Nov. 2024). DOI: [10.48550/arXiv.2411.19908](https://doi.org/10.48550/arXiv.2411.19908).
- [3] V Kanwar et al. “Multi-Artifact Detection and Filtering in Digital Pathology Using Intrinsic Image Properties”. In: *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*. Aug. 2024, pp. 1–5. DOI: [10.1109/ISBI56570.2024.10635902](https://doi.org/10.1109/ISBI56570.2024.10635902).
- [4] A Woicik et al. “In Silico Optimization of Tissue Microarray Design for Machine Learning Analysis”. In: *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*. Aug. 2024, pp. 1–5. DOI: [10.1109/ISBI56570.2024.10635208](https://doi.org/10.1109/ISBI56570.2024.10635208).
- [5] ZR McCaw et al. “Machine learning enabled prediction of digital biomarkers from whole slide histopathology images”. In: *medRxiv* (Jan. 2024). DOI: [10.1101/2024.01.06.24300926](https://doi.org/10.1101/2024.01.06.24300926).
- [6] C Angermueller, Z Mariet, B Jester, et al. “High-throughput ML-guided design of diverse single-domain antibodies against SARS-CoV-2”. In: *bioRxiv* (Dec. 2023). DOI: [10.1101/2023.12.01.569227](https://doi.org/10.1101/2023.12.01.569227).
- [7] ZR McCaw, H Julienne, and H Aschard. “Fitting Gaussian mixture models on incomplete data”. In: *BMC Bioinformatics* 23.1 (June 2022), p. 208. DOI: [10.1186/s12859-022-04740-9](https://doi.org/10.1186/s12859-022-04740-9).

Biostatistics & Clinical Trials

- [1] EJ Hsu et al. “Differential censoring in overall survival in phase 3 oncology clinical trials”. In: *Journal of the National Cancer Institute* djaf378 (Dec. 2025). DOI: [10.1093/jnci/djaf378](https://doi.org/10.1093/jnci/djaf378).
- [2] S Armbruster, ZR McCaw, K Jering, et al. “Capturing Totality of Treatment Effects: A Model-Free Approach to Analyzing Multiple Event-Time Data From Heart Failure Studies”. In: *JACC Heart Failure* (Aug. 2025). DOI: [10.1016/j.jchf.2025.102605](https://doi.org/10.1016/j.jchf.2025.102605).

- [3] AD Sherry, P Msaouel, AM Miller, et al. “Reproducibility of statistically significant phase III oncology trials: An In Silico meta-epidemiological analysis”. In: *European Journal of Cancer* (Aug. 2025). DOI: [10.1016/j.ejca.2025.115596](https://doi.org/10.1016/j.ejca.2025.115596).
- [4] AD Sherry et al. “Bayesian Interim Analysis and Efficiency of Phase III Randomized Trials”. In: *Br J Cancer* (Aug. 2025). DOI: [10.1038/s41416-025-03156-5](https://doi.org/10.1038/s41416-025-03156-5).
- [5] AD Sherry, Y Liu, P Msaouel, et al. “Survival-inferred fragility of statistical significance in phase III oncology trials”. In: *NPJ Precision Oncology* (July 2025). DOI: [10.1038/s41698-025-01024-2](https://doi.org/10.1038/s41698-025-01024-2).
- [6] TJ Kleber, AD Sherry, AJ Arifin, et al. “Justification, margin values, and analysis populations for oncologic noninferiority and equivalence trials: a meta-epidemiological study”. In: *Journal of the National Cancer Institute* (May 2025). DOI: [10.1093/jnci/djae318](https://doi.org/10.1093/jnci/djae318).
- [7] J Gronsbell et al. “Exact Inference for Random Effects Meta-Analyses for Small, Sparse Data”. In: *Stats* 8.1 (Mar. 2025). DOI: [10.3390/stats8010005](https://doi.org/10.3390/stats8010005).
- [8] AD Sherry et al. “Evidence-Based Prior for Estimating the Treatment Effect of Phase III Randomized Trials in Oncology”. In: *JCO Precision Oncology* 8 (Oct. 2024), e2400363. DOI: [10.1200/PO.24.00363](https://doi.org/10.1200/PO.24.00363).
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- [10] AD Sherry et al. “Off-Protocol Radiation Therapy in Phase 3 Metastatic Solid Tumor Trials”. In: *Int J Radiat Oncol Biol Phys* (Aug. 2024). DOI: [10.1016/j.ijrobp.2024.05.033](https://doi.org/10.1016/j.ijrobp.2024.05.033).
- [11] AD Sherry et al. “Improving the clinical meaning of surrogate endpoints: An empirical assessment of clinical progression in phase III oncology trials”. In: *International Journal of Cancer* (Aug. 2024). DOI: [10.1002/ijc.35129](https://doi.org/10.1002/ijc.35129).
- [12] TA Lin et al. “Proportional Hazards Violations in Phase 3 Cancer Clinical Trials: A Potential Source of Trial Misinterpretation”. In: *Clinical Cancer Research* (Aug. 2024). DOI: [10.1158/1078-0432.CCR-24-0566](https://doi.org/10.1158/1078-0432.CCR-24-0566).
- [13] AD Sherry et al. “Towards Treatment Effect Interpretability: A Bayesian Re-analysis of 194,129 Patient Outcomes Across 230 Oncology Trials”. In: *medRxiv* (July 2024). DOI: [10.1101/2024.07.23.24310891](https://doi.org/10.1101/2024.07.23.24310891).
- [14] EJ Hsu et al. “Association of differential censoring with survival and suboptimal control arms among oncology clinical trials”. In: *Journal of the National Cancer Institute* 116.6 (June 2024), pp. 990–994. DOI: [10.1093/jnci/djae028](https://doi.org/10.1093/jnci/djae028).
- [15] AD Sherry, P Msaouel, TA Lin, et al. “Postprogression therapy and confounding for the estimated treatment effect on overall survival in phase III oncology trials”. In: *BMJ Oncology* (Apr. 2024). DOI: [10.1136/bmjonc-2024-000322](https://doi.org/10.1136/bmjonc-2024-000322).
- [16] AD Sherry, AW Hahn, ZR McCaw, et al. “Differential Treatment Effects of Subgroup Analyses in Phase 3 Oncology Trials From 2004 to 2020”. In: *JAMA Network Open* 7.3 (Mar. 2024), e243379. DOI: [10.1001/jamanetworkopen.2024.3379](https://doi.org/10.1001/jamanetworkopen.2024.3379).

- [17] AD Sherry, P Msaouel, ZR McCaw, et al. “Prevalence and implications of significance testing for baseline covariate imbalance in randomised cancer clinical trials: The Table 1 Fallacy”. In: *European Journal of Cancer* 194 (Nov. 2023), p. 113357. DOI: [10.1016/j.ejca.2023.113357](https://doi.org/10.1016/j.ejca.2023.113357).
- [18] X Wang et al. “Using a Clinically Interpretable End Point Composed of Multiple Outcomes to Evaluate Totality of Treatment Effect in Comparative Oncology Studies”. In: *JAMA Network Open* 6.6 (June 2023), e2319055. DOI: [10.1001/jamanetworkopen.2023.19055](https://doi.org/10.1001/jamanetworkopen.2023.19055).
- [19] PS Jhund et al. “Effect of Dapagliflozin on Total Heart Failure Events in Patients With Heart Failure With Mildly Reduced or Preserved Ejection Fraction: A Prespecified Analysis of the DELIVER Trial”. In: *JAMA Cardiology* (Apr. 2023). DOI: [10.1001/jamacardio.2023.0711](https://doi.org/10.1001/jamacardio.2023.0711).
- [20] A Das et al. “Assessment of Median and Mean Survival Time in Cancer Clinical Trials”. In: *JAMA Network Open* 6.4 (Apr. 2023), e236498. DOI: [10.1001/jamanetworkopen.2023.6498](https://doi.org/10.1001/jamanetworkopen.2023.6498).
- [21] HM Dehbi, A Embleton-Thirsk, and McCaw ZR. “Sample size calculation for randomized selection trials with a time-to-event endpoint and a margin of practical equivalence”. In: *Statistics in Medicine* (June 2022). DOI: [10.1002/sim.9490](https://doi.org/10.1002/sim.9490).
- [22] BL Claggett et al. “Quantifying Treatment Effects in Trials with Multiple Event-Time Outcomes”. In: *NEJM Evidence* 1.10 (June 2022). DOI: [10.1056/EVIDoa2200047](https://doi.org/10.1056/EVIDoa2200047).
- [23] ZR McCaw, DH Kim, and LJ Wei. “Pitfall in the Design and Analysis of Comparative Oncology Trials With a Time-to-Event Endpoint and Recommendations”. In: *JNCI Cancer Spectrum* 6.1 (Feb. 2022), pkac007. DOI: [10.1093/jncics/pkac007](https://doi.org/10.1093/jncics/pkac007).
- [24] ZR McCaw et al. “Practical Recommendations on Quantifying and Interpreting Treatment Effects in the Presence of Terminal Competing Risks: A Review”. In: *JAMA Cardiology* (Dec. 2021). DOI: [10.1001/jamacardio.2021.4932](https://doi.org/10.1001/jamacardio.2021.4932).
- [25] ZR McCaw et al. “Choosing clinically interpretable summary measures and robust analytic procedures for quantifying the treatment difference in comparative clinical studies”. In: *Statistics in Medicine* 40.28 (Dec. 2021), pp. 6235–6242. DOI: [10.1002/sim.8971](https://doi.org/10.1002/sim.8971).
- [26] R Sun et al. “Moving beyond conventional stratified analysis to assess the treatment effect in a comparative oncology study”. In: *Journal for ImmunoTherapy of Cancer* 9.11 (Nov. 2021), e003323. DOI: [10.1136/jitc-2021-003323](https://doi.org/10.1136/jitc-2021-003323).
- [27] ZR McCaw et al. “Neoadjuvant chemotherapy in bladder cancer: Clinical benefit observed in prospective trials computed with restricted mean survival times”. In: *Urologic Oncology* S1078-1439.20 (Jan. 2021), pp. 30640–30642. DOI: [10.1016/j.urolonc.2020.12.012](https://doi.org/10.1016/j.urolonc.2020.12.012).
- [28] ZR McCaw et al. “Survival analysis of treatment efficacy in comparative COVID-19 studies”. In: *Clinical Infectious Diseases* (Oct. 2020). DOI: [10.1093/cid/ciaa1563](https://doi.org/10.1093/cid/ciaa1563).

- [29] C Perego et al. “Utility of Restricted Mean Survival Time Analysis for Heart Failure Clinical Trial Evaluation and Interpretation”. In: *JACC Heart Failure* (Oct. 2020). DOI: [10.1016/j.jchf.2020.07.005](https://doi.org/10.1016/j.jchf.2020.07.005).
- [30] ZR McCaw et al. “Selecting Appropriate Endpoints for Assessing Treatment Effects in Comparative Clinical Studies for COVID-19”. In: *Contemporary Clinical Trials* (Sept. 2020). DOI: [10.1016/j.cct.2020.106145](https://doi.org/10.1016/j.cct.2020.106145).
- [31] ZR McCaw et al. “How to Quantify and Interpret Treatment Effects in Comparative Clinical Studies of COVID-19”. In: *Annals of Internal Medicine* (July 2020). DOI: [10.7326/M20-4044](https://doi.org/10.7326/M20-4044).
- [32] B Huang et al. “Analysis of Response Data for Assessing Treatment Effects in Comparative Clinical Studies”. In: *Annals of Internal Medicine* (July 2020). DOI: [10.7326/M20-0104](https://doi.org/10.7326/M20-0104).
- [33] ZR McCaw, G Yin, and LJ Wei. “Using the Restricted Mean Survival Time Difference as an Alternative to the Hazard Ratio for Analyzing Clinical Cardiovascular Studies”. In: *Circulation* 140.17 (Oct. 2019), pp. 1366–1368. DOI: [10.1161/CIRCULATIONAHA.119.040680](https://doi.org/10.1161/CIRCULATIONAHA.119.040680).
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Biology

- [1] L Siraj et al. “Functional dissection of complex trait variants at single-nucleotide resolution”. In: *Nature* (Feb. 2026). DOI: [10.1038/s41586-026-10121-6](https://doi.org/10.1038/s41586-026-10121-6).
- [2] J Marzec et al. “Toll-like receptor 4-mediated respiratory syncytial virus disease and lung transcriptomics in differentially susceptible inbred mouse strains”. In: *Physiological Genomics* (Nov. 2019). DOI: [10.1152/physiolgenomics.00101.2019](https://doi.org/10.1152/physiolgenomics.00101.2019).
- [3] M High et al. “Determinants of host susceptibility to murine respiratory syncytial virus (RSV) disease identify a role for the innate immunity scavenger receptor MARCO gene in human infants”. In: *EBioMedicine* S2352-3964.16 (2016), pp. 30360–7. DOI: [10.1016/j.ebiom.2016.08.011](https://doi.org/10.1016/j.ebiom.2016.08.011).
- [4] JM Ciencewicki et al. “Effects of mannose-binding lectin on pulmonary gene expression and innate immune inflammatory response to ozone”. In: *American Journal of Physiology-Lung Cellular and Molecular Physiology* 311.2 (2016), pp. 280–91. DOI: [10.1152/ajplung.00205.2015](https://doi.org/10.1152/ajplung.00205.2015).
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- [6] KC Verhein et al. “Novel Roles for Notch3 and Notch4 Receptors in Gene Expression and Susceptibility to Ozone Induced Lung Inflammation in Mice”. In: *Environmental Health Perspectives* 123.8 (2015), pp. 799–805. DOI: [10.1289/ehp.1408852](https://doi.org/10.1289/ehp.1408852).

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- [9] KC Verhein et al. “Genetic Factors Involved in Susceptibility to Lung Disease”. In: *The Lung Second Edition: Development, Aging and the Environment*. Ed. by Plopper CG Harding R Pinkerton KE. London: Academic Press, 2014. DOI: [10.1016/B978-0-12-799941-8.00020-1](https://doi.org/10.1016/B978-0-12-799941-8.00020-1).

Correspondence

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Professional Activities

- **ML4H Section Chair** 2025
- **Peer Review** 2025
Journals: Bioinformatics, BMC Bioinformatics, BMC Medical Research Methodology, Cancers, Clinical Trials, EJMR, iScience, ISMB, Nature Aging, Nutrition, Nature Communications, npj Health Systems
- **Peer Review** 2024
Journals: Bioinformatics, Briefings in Bioinformatics, BMC Medical Research Methodology, ISCB, ISMB, Statistics in Medicine, Stats
- **Peer Review** 2023
Journals: Current Cancer Drug Targets, ISCB, RECOMB
- **Peer Review** 2022
Journals: Axioms, ISCB, Life, Statistics in Biopharmaceutical Research, Statistics in Medicine, TEST, Viruses
- **Peer Review** 2021
Journals: Circulation: Cardiovascular Quality and Outcomes, Frontiers in Genetics, ISCB, Statistics in Medicine
- **Peer Review** 2020
Journals: ISCB, Statistics in Medicine

- **JSM Section Chair** 2019
Regression Methods for Longitudinal Data
- **JSM Section Chair** 2018
Gene-Gene and Gene-Environment Interactions

Conference Presentations

- **American Society of Human Genetics** 11/2024
Unveiling the power of allelic series: enhancements and applications of COAST.
- **American Association for Cancer Research** 04/2024
Machine learning enabled prediction of digital biomarkers from whole slide histopathology images.
- **American Society of Human Genetics** 10/2023
Synthetic slope analysis for progression GWAS.
- **American Association for Cancer Research** 04/2023
Learned phenotypic embeddings enable scalable imputation of high-content molecular data elucidating prognostic chromatin signatures.
- **American Society of Human Genetics** 10/2022
An allelic series rare variant association test for candidate gene discovery.
- **American Society of Human Genetics** 10/2019
Cross-population fine-mapping to identify shared and population-specific causal effects.
- **Joint Statistical Meeting** 07/2019
Cross-tissue eQTL calling via surrogate expression analysis.
- **Harvard School of Public Health, Program in Quantitative Genomics** 11/2018
Leveraging the UKB to empower association testing on scarce phenotypes.
- **Joint Statistical Meeting** 07/2018
Leveraging surrogate phenotypes to improve inference on a partially missing target phenotype.
- **Joint Statistical Meeting** 07/2017
Inverse normal transformation for genome-wide association testing of quantitative traits.
- **American Thoracic Society** 05/2014
Gene expression profiling predicts response to respiratory syncytial virus (RSV) in mice.
- **National Institute of Environmental Health Sciences** 07/2011
Identifying candidate susceptibility genes for respiratory syncytial virus (RSV) disease severity.
- **National Institute of Environmental Health Sciences** 07/2010
Characterization of transcriptional networks underlying Tlr4-mediated respiratory syncytial virus (RSV) disease in mice.

Awards and Distinctions

- **Distinguished Student Paper Award** 07/2019
Joint Statistical Meeting, Section in Genetics and Genomics.
- **Stellar Abstract Award** 11/2018
Harvard School of Public Health, Program in Quantitative Genomics
- **Ruth L. Kirschstein National Research Service Award (F31)** 03/2018
[Innovations in Genome Wide Association Testing Inspired by Obstructive Sleep Apnea Phenotypes](#)
- **Teaching Fellow** 11/2017
Global Initiative for Neuropsychiatric Genetic Education in Research
- **NIH Pre-Doctoral Training Grant** 08/2016
Statistical and Quantitative Training in Big Data Health Science
- **NIH Pre-Doctoral Training Grant** 08/2014
Interdisciplinary Training Grant in Biostatistics and Computational Biology
- **NIH Post-Baccalaureate Research Fellow** 09/2013
National Institute of Environmental Health Sciences
- **Undergraduate Academic Achievement Award** 04/2013
UNC Department of Biostatistics
- **Phi Beta Kappa National Honors Society** 11/2011
- **NIH Summer Internship** 05/2011
National Institute of Environmental Health Sciences 05/2010

Teaching Experience

UNC CHAPEL HILL 06/2023 – Present
ADJUNCT INSTRUCTOR

- DEPARTMENT: MPH@UNC
- CLASSES: Section instructor for SPHG 711, Data Analysis for Public Health.

UNC CHAPEL HILL

- CLASS: Data Analysis for Public Health (SPHG 711) 08/2024 - 12/2024
08/2023 - 12/2023

HARVARD UNIVERSITY

- CLASS: Inference II (BST 241) 02/2019 – 05/2019
INSTRUCTOR: Rui Wang, Ph.D.
- CLASS: Introduction to Biostatistics 02/2019
INSTRUCTOR: Lori Chibnik, Ph.D.
LOCATION: University of KwaZulu-Natal, Durban, South Africa

- CLASS: Multivariate and Longitudinal Analysis (BST 245) 02/2018 – 05/2018
INSTRUCTOR: Sebastien Haneuse, Ph.D.
- CLASS: Inference I (BST 231) 02/2017 – 05/2017
INSTRUCTOR: Judith Lok, Ph.D.
- CLASS: Statistical Genetics (BST 227) 10/2016 – 12/2016
INSTRUCTOR: Martin Aryee, Ph.D.
- CLASS: Computational Biology (STAT 215) 02/2016 – 05/2016
INSTRUCTOR: X. Shirley Liu, Ph.D.

UNC CHAPEL HILL

- CLASS: General Chemistry I (CHEM 101) 08/2012 – 12/2012
INSTRUCTOR: Jennifer Krumper, Ph.D.
- CLASS: Organic Chemistry II (CHEM 262) 08/2011 – 12/2011
INSTRUCTOR: Jennifer Krumper, Ph.D.